

WHY ARE WE ASKING THIS QUESTION?

The Question Behind Every ESG Disclosure

Why "is this an ESG issue" and "is this financially material" are not the same question.

Sustainability related disclosure has expanded enormously since 2023: IFRS S1 and IFRS S2 now require entities to report on sustainability related risks and opportunities that could reasonably affect the entity's cash flows, access to finance, or cost of capital. ■ **OBSERVED FACT**

But a disclosure requirement is not the same thing as a claim that every disclosed issue moves the numbers. Two propositions are routinely collapsed into one, and this dossier exists to pull them apart.

PROPOSITION OFTEN ASSUMED

"This is an ESG issue, therefore it affects company value."

PROPOSITION THIS DOSSIER TESTS

"This is an ESG issue. Does it, how, how much, and under what assumptions does it affect company value?"

Coca-Cola's FY2025 Form 10-K discusses water availability and stewardship, but the discussion sits in Item 1 (Business), describing operational context, not in Item 1A (Risk Factors), where the company flags matters it considers a material risk to results. The same topic can be strategically relevant without meeting a company's own bar for a disclosed material risk.

■ **OBSERVED FACT**

This distinction is not academic. Modules 2 through 7 exist specifically to test it — page by page, mechanism by mechanism — against real disclosures, real financial statements, and real capital allocation decisions, rather than against the assumption that disclosure and materiality are the same fact.

THE CENTRAL QUESTION OF THIS STUDY

When does a sustainability related issue become financially material to a company?

Not: does ESG create value. Not: does ESG destroy value. This dossier asks when, through what mechanism, at what magnitude, under which assumptions, and with what management and capital allocation consequences an issue changes the economics of a company.

WHY MISTAKING THE TWO IS COSTLY

False positive. Treating a disclosed topic as material risk with no identifiable transmission mechanism, diluting analytical attention.

False negative. Dismissing a topic as "just ESG" when it already sits inside cost of capital, capex, or asset values.

Both errors are avoidable with one discipline: trace the mechanism before concluding materiality.

HOW THIS DOSSIER SIGNALS RIGOR

Every substantive claim below is tagged by evidence type — observed fact, management claim, analyst inference, or analytical synthesis — so the reader always knows which kind of statement they are reading, and can weigh it accordingly.

THE ROUTE THIS DOSSIER TAKES

01–03 Foundations What the two words actually mean	04–06 Transmission How an issue enters the model	07–09 Magnitude & exposure How large, how certain, to whom	10–12 Identify & prioritize From a list to a ranked set	13–15 Management & capital What a company can do, and its cost	16–18 Uncertainty & decision Scenarios, stress, strategy	19–20 Integration One case, one framework
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WHAT DOES FINANCIAL MATERIALITY MEAN?

What Financial Materiality Actually Means

The formal test: nature, magnitude, and one company's specific facts and circumstances.

THE FORMAL DEFINITION, ISSB

Sustainability related financial information is material if omitting, misstating, or obscuring that information could reasonably be expected to influence decisions that primary users of general purpose financial reports make on the basis of those reports.

IFRS S1, §18 (issued June 2023), adapted from the equivalent financial materiality test in IAS 1.

■ OBSERVED FACT

THE MATERIALITY TEST HAS TWO MOVING PARTS

Nature. Qualitative character of the item: a regulatory breach, a related party matter, or a going concern indicator can be material regardless of size.

Magnitude. Quantitative size of the effect relative to the entity's financial position and performance.

IFRS Practice Statement 2, *Making Materiality Judgements*: a facts and circumstances test, not a fixed percentage threshold. ■ OBSERVED FACT

SAME TOPIC, OPPOSITE CONCLUSION: ILLUSTRATIVE CONTRAST

TOPIC	LIKELY MATERIAL CASE	UNLIKELY MATERIAL CASE
Water scarcity	Beverage bottler in a stressed basin	Software company, negligible water use
Carbon pricing	Cement producer, high direct emissions	Consulting firm, minimal direct emissions
Supply chain labour	Apparel brand, complex tiered supply chain	Domestic only software vendor

■ ANALYTICAL SYNTHESIS Illustrative contrasts constructed to demonstrate the entity specific principle, not disclosed company positions.

KEY TAKEAWAY

Materiality is a test applied to one company at a time. A topic being an ESG issue establishes candidacy for the test. It does not pass it automatically.

Sources: [IFRS Foundation — IFRS Practice Statement 2, Making Materiality Judgements](#) · [SEC — Staff Accounting Bulletin No. 99](#) (parallel only, not an ISSB source)

ENTITY SPECIFIC, NOT ISSUE SPECIFIC

Materiality attaches to the combination of an issue and a specific company's facts and circumstances, not to the issue in the abstract. The same topic, water scarcity for example, can be material for a beverage bottler in a water stressed basin and immaterial for a software company with negligible water use.

A NON ISSB PARALLEL, LABELLED AS SUCH

US securities law applies an analogous "no bright line numerical threshold" logic to financial statement materiality generally.

SEC Staff Accounting Bulletin 99, used here only as a labelled parallel illustrating that qualitative, facts and circumstances materiality tests are not unique to sustainability reporting. This is not an ISSB source and is not conflated with one. ■ OBSERVED FACT

HOW CAN WE FORMALIZE IT?

Five Words That Are Not Synonyms

Financial materiality, impact materiality, double materiality, investor materiality, and ESG score, formally distinguished.

CONCEPT	TEST APPLIED	GOVERNING FRAMEWORK	STATUS HERE
Financial materiality	Could this issue reasonably affect the entity's own cash flows, access to finance, or cost of capital?	IFRS S1 / S2 (ISSB)	■ OBSERVED FACT
Impact materiality	Does the entity's activity have a significant effect on people or the environment, regardless of financial effect?	ESRS (EU CSRD)	■ OBSERVED FACT
Double materiality	Union of financial and impact materiality; an issue is reported if either test is met	ESRS (EU CSRD)	■ OBSERVED FACT
Investor materiality	Would a reasonable investor's portfolio decision change on this information?	CFA Institute usage	■ OBSERVED FACT
ESG score	A third party rating provider's composite index; its own methodology, not a financial materiality judgement	Rating providers, not a standard setter	■ NOT EQUIVALENT

WHY THE ESRS / ISSB DISTINCTION MATTERS HERE

A joint ISSB, European Commission, and EFRAG interoperability statement, 2 May 2024, is the strongest available source confirming that ISSB (financial materiality, single) and ESRS (double materiality) are related but non identical constructs designed to be used together, not interchangeably. ■ OBSERVED FACT

WHAT SASB CONTRIBUTES, AND WHAT IT DOES NOT

SASB industry based topics, now part of the IFRS Foundation, help identify candidate issues for a given industry. IFRS S1 requires an entity to consider SASB; it does not make SASB topics mandatorily material. Direct navigator access was not obtainable during this project; company published SASB indices, Coca-Cola HBC and IBM among them, were used instead.

■ COMPANY PUBLISHED INDEX, GRADE B

Financial	Impact + Financial	≠
The test this dossier applies	ESRS "double" superset	ESG score is not a materiality test

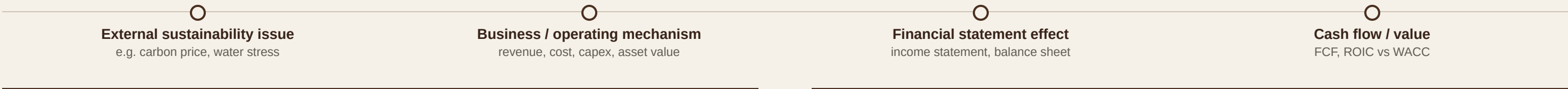
MODULE 1 CONCLUSION

A topic can be an SASB listed industry topic, an ESRS impact material issue, and a low ESG score, without ever clearing the financial materiality bar this dossier tracks. The remaining seventeen pages trace the mechanism that decides which topics do.

THROUGH WHAT MECHANISM DOES THE ISSUE AFFECT THE BUSINESS?

From "Does It Matter" to "How Does It Enter the Model"

The financial transmission chain: external issue, operating mechanism, financial statement effect, cash flow.



FIVE INTEGRATION METHODS INTO VALUATION

CFA Institute identifies five ways ESG factors are translated into standard financial analysis variables: adjusting revenue growth, margins, required investment, the discount rate, or applying scenario analysis. The unifying principle: an ESG issue is analysed through the existing valuation model, not appended as a separate line item. ■ **OBSERVED FACT**

CFA Institute, ESG Integration Guidance, Nov 2022.

IMPORTANT SCOPE NOTE

No single standard setter publishes one unified mechanical formula linking an external driver to free cash flow. The chain above synthesises TCFD's disclosure categories with standard accounting identities. ■ **ANALYTICAL SYNTHESIS**

THE FIVE CFA INTEGRATION METHODS, APPLIED TO A WORKING EXAMPLE

Revenue growth	Margins	Required investment	Discount rate	Scenario analysis
Build a demand or pricing effect into top line assumptions, e.g. low carbon product premium.	Adjust cost assumptions for input price or compliance cost changes.	Raise or lower planned capex to reflect adaptation or compliance spend.	Adjust WACC for perceived risk change, used cautiously; effects are often better modelled in cash flows.	Run the model across more than one plausible future rather than picking one point estimate.

■ **ANALYTICAL SYNTHESIS** Descriptions of each method are this dossier's own elaboration of the CFA Institute categories; the five category names themselves are the cited source's own taxonomy.

KEY TAKEAWAY

A mechanism is a claim about a specific causal path from issue to cash flow, not a general association. Pages 5 and 6 test that path against real financial statements and a difficult real world example.

Sources: [CFA Institute — Guidance for Integrating ESG Information into Equity Analysis \(Nov 2022\)](#) · [TCFD — Technical Supplement, Fig. 3 \(Jun 2017\)](#)

WHAT DOES THE MECHANISM LOOK LIKE FINANCIALLY?

Where the Effect Actually Lands

Mapping revenue, cost, capex, asset, and financing channels onto the three financial statements.

REVENUE CHANNEL

Customer demand shifts, low carbon product preference for example, regulatory market access, or pricing power tied to a sustainability attribute. Effect shows up first in the income statement top line.

COST & CAPEX CHANNEL

Input price volatility, compliance cost, insurance and reinsurance repricing, and capital expenditure required to adapt operations or comply with regulation.

ASSET & FINANCING CHANNEL

Impairment of carrying values when future cash flows no longer support them, and changes in the cost or availability of debt and equity capital tied to perceived risk.

WHERE EACH CHANNEL LANDS IN THE FINANCIAL STATEMENTS

CHANNEL	INCOME STATEMENT	BALANCE SHEET	CASH FLOW
Revenue	Top line, margin mix	None	Operating CF
Cost	COGS / opex, margins	Working capital	Operating CF
Capex	Depreciation (lagged)	PP&E, capital employed	Investing CF
Assets	Impairment charge	Carrying value ↓	Non cash
Financing	Interest expense	Debt / equity mix	Financing CF

■ **ANALYTICAL SYNTHESIS** Category structure follows TCFD Fig. 3 output categories combined with standard IFRS financial statement elements, not one single authoritative table.

WHY MULTIPLE CHANNELS AT ONCE MATTERS

Analysts who check only one channel, typically revenue, can miss the larger effect sitting in capex or asset carrying values. A rigorous transmission analysis checks all five channels before concluding an issue is immaterial.

A NOTE ON DIRECTION

Nothing about the five channel framework presumes the effect is negative. The same channels carry cost savings, revenue gains, and asset appreciation exactly as readily as their opposites. Direction is an empirical question, not a framework assumption.

KEY TAKEAWAY

The same external issue can travel through more than one channel simultaneously. The next page shows one issue producing several distinct financial effects in a single company.

CAN WE SEE THE MECHANISM IN A REAL, DIFFICULT EXAMPLE?

One Mechanism That Creates Value, One That Destroys It

Tracing the same five channel logic through a real cost saving case and a real impairment case.

UPS — ORION Route Optimization

Cost channel · value created

ISSUE	Fuel consumption and delivery mile inefficiency across the US driver fleet.
MECHANISM	Algorithmic route optimization (ORION) deployed fleet wide, decoupling miles driven from package volume.
EVIDENCE	2016: daily volume & stops up 4.1% / 4.4%; daily miles driven up only 0.2%. ■ OBSERVED FACT
FINANCIAL EFFECT	Company states more than \$400m in annual cost savings and avoidance. ■ OBSERVED FACT
INTERPRETATION	A disclosed mechanism, not just an assertion: the operating fact (miles vs. volume) and the financial claim move together.
LIMITATION	The \$250m deployment cost is INFORMS' figure, not UPS's own filing, and is a decade old. ■ THIRD PARTY ESTIMATE

THE UPS MECHANISM, TRACED END TO END



WHICH OF THE FIVE CHANNELS (PAGE 5) DOES EACH CASE ACTUALLY EXERCISE?

CHANNEL	UPS ORION	ØRSTED US OFFSHORE
Revenue	Not the primary channel	Indirect, via future contracted output
Cost	Primary — fuel and labour	Secondary
Capex	Technology deployment cost	Primary — construction spend
Assets	Not applicable	Primary — impairment
Financing	Not disclosed	Cost of capital / dividend pause

Ørsted — US Offshore Wind

Asset & capex channel · value destroyed

ISSUE	US permitting delay, financing cost increases, and supply chain inflation on Ocean Wind 1/2.
MECHANISM	Capitalised construction costs no longer supported by expected future cash flows.
EVIDENCE	FY2023 audited impairment of DKK 26.8bn ; an earlier nine month figure of DKK 28.4bn does not reconcile with it. ■ OBSERVED FACT
FINANCIAL EFFECT	Development of Ocean Wind 1 & 2 ceased; Revolution Wind's FID taken the same day. ■ OBSERVED FACT
INTERPRETATION	Developed at transmission level only; whether this destroyed economic value against the cost of capital is tested on pages 14–15. ■ ANALYTICAL SYNTHESIS
LIMITATION	The two impairment figures are shown as disclosed, not silently reconciled. ■ EVIDENCE GAP

The transmission mechanism can be traced with primary evidence in both directions — a cost saving mechanism and a value destroying one. Neither outcome is assumed; both are shown because the evidence supports them.

HOW LARGE COULD THE EFFECT BE?

How Large Could the Effect Be?

Magnitude is measured relative to the entity's own financial position, never as an absolute figure.

IFRS Practice Statement 2 tests magnitude relative to the entity's own financial position and performance, not against an absolute currency threshold and not against industry norms. A DKK 1bn effect is immaterial to a company with a DKK 500bn balance sheet, and potentially decisive for one a tenth that size. ■ **OBSERVED FACT**

SYSTEM LEVEL MAGNITUDE

The ECB's 2022 supervisory climate stress test found aggregate credit and market losses of around €70 billion across 41 banks under disorderly transition and physical risk scenarios, a system level, not single company, magnitude reference.

ECB Banking Supervision, 2022 Climate Risk Stress Test (8 Jul 2022). ■ **SYSTEM LEVEL EVIDENCE**

WHY THIS MATTERS FOR AN ANALYST

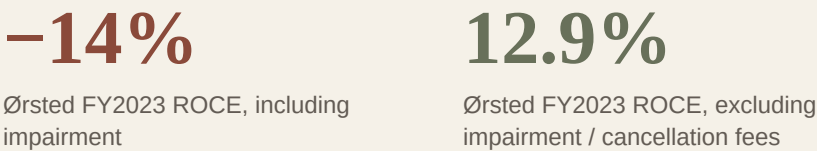
A magnitude figure without a denominator is not yet informative. "€70bn of losses" and "DKK 26.8bn of impairment" only become analytically useful once scaled against the relevant balance sheet, capital base, or sector total, which is why this dossier never quotes a magnitude figure without also stating what it is measured against.

THE SAME PAIR OF NUMBERS, VISUALISED: ØRSTED FY2023 ROCE, %



Chart is a direct visualisation of the two disclosed figures, not a calculated or estimated series. ■ **OBSERVED FACT**

COMPANY LEVEL MAGNITUDE, SHOWN NOT ASSERTED



Both figures must be read together: the headline number shows the magnitude of the write down; the underlying number shows that operating profitability was not simultaneously destroyed. ■ **OBSERVED FACT**

KEY TAKEAWAY

Magnitude is relative, not absolute. The next page adds the second dimension, how certain, and over what time horizon, before page 9 asks why identical magnitude can still produce different outcomes at different companies.

HOW UNCERTAIN IS IT?

How Certain, and Over What Horizon?

Likelihood, time horizon, and outcome range are three separate dimensions of uncertainty, not one.

UNCERTAINTY IS A RANGE, NOT A GUESS

Both IFRS S2 and NGFS treat the future as a range of plausible outcomes, explicitly not a forecast.

"The NGFS scenarios are not forecasts. They aim at exploring plausible futures, neither necessarily the most probable nor the most desirable."

NGFS, Climate Scenarios Technical Documentation, Phase V (Nov 2024), p.7. ■ **OBSERVED FACT** IFRS S2 defines scenario analysis as identifying "a potential range of outcomes of future events under conditions of uncertainty." ■ **OBSERVED FACT**

HOW NGFS ORGANISES ITS OWN RANGE OF PLAUSIBLE FUTURES

Orderly	Disorderly	Hot house world	Too little, too late
Early, gradual policy action	Late, abrupt policy action	Limited additional action; physical risk dominates	Insufficient, delayed action on both fronts

■ **OBSERVED FACT** NGFS scenario families, Technical Documentation, Phase V (Nov 2024): four named, internally consistent alternative states, not four probability ranked forecasts.

WHERE THIS GOES NEXT IN THE DOSSIER

The full mechanics of scenario construction, stress testing, and reverse stress testing are developed in Module 6, pages 16 to 18, once management response and capital allocation, Module 5, have been established. Introducing them here would be premature; this page fixes only the vocabulary: uncertainty is a described range with a time horizon attached, not a single number.

■ ANALYTICAL SYNTHESIS

KEY TAKEAWAY

A magnitude estimate without a stated likelihood and time horizon is incomplete. Page 9 now asks why two companies facing the identical external driver, at the identical magnitude and uncertainty, can still end up with very different financial consequences.

WHY DOES COMPANY EXPOSURE MATTER?

Why Company Exposure Is Not Optional Detail

Exposure, vulnerability, and resilience explain why identical external issues produce different company level outcomes.

Exposure

Is the company's activity physically or economically in the path of the driver at all, e.g. asset location, revenue concentration?

Vulnerability

Given exposure, how sensitive are this company's specific economics: margin structure, contract terms, substitutability?

Resilience

Given vulnerability, what buffers exist: balance sheet strength, diversification, pricing power, insurance?

■ **ANALYTICAL SYNTHESIS** This three part decomposition is the dossier's own organising structure for company specific exposure analysis, built to keep "is the company in the path of the risk" analytically separate from "can the company absorb it."

ØRSTED, REVISITED FOR EXPOSURE, NOT OUTCOME

Within one company, exposure itself varied by geography: the US offshore pipeline, Ocean Wind 1/2, proved materially more exposed to permitting delay, US interest rate movements, and supply chain inflation than the European portfolio in the same period. Same parent company, same core strategy, different geographic exposure, and a different outcome.

■ **OBSERVED FACT** Ørsted, Oct 2023 announcement. Used here for its exposure lesson; the capital allocation and value creation reading of the same case is developed separately on pages 13 to 15.

THE THREE FILTERS APPLIED IN SEQUENCE



In the path of the driver, at all? Then how sensitive, given exposure? Then what buffers exist? Then what actually reaches the numbers.

WHY THIS IS NOT REDUNDANT WITH MAGNITUDE

Two insurers can carry an identical dollar exposure to a peril such as wildfire or coastal flood risk and still reach opposite underwriting conclusions, because vulnerability, concentration, rate filing constraints, and resilience, reinsurance access, diversification, capital, differ company by company. Exposure alone never tells the whole story.

MODULE 3 CONCLUSION

Exposure, vulnerability, and resilience are analytically distinct. Module 4 now moves from analysing one company at a time to identifying and comparing issues across an industry and its peers.

HOW DO WE IDENTIFY RELEVANT ISSUES?

Identifying Candidate Issues by Industry

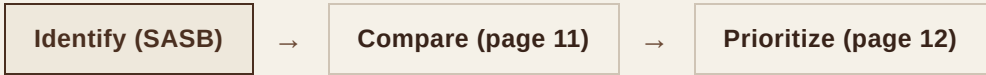
SASB topics narrow an unbounded issue universe to an industry relevant shortlist. They do not decide materiality.

SASB: A STARTING LIST, NOT A VERDICT

SASB industry based topics, now maintained by the IFRS Foundation, identify which sustainability topics are reasonably likely to be financially relevant for a given industry, a candidate list built from industry wide research, not a company specific materiality determination.

IFRS S1 requires an entity to consider the SASB Standards applicable to its industry. The standard does not make every listed topic mandatorily material for every company in that industry. ■ **OBSERVED FACT**

IDENTIFICATION IS THE FIRST STEP OF THREE



Identify industry based candidate topics, then compare through peer and industry benchmarking, then prioritize through materiality mapping for one company. ■ **ANALYTICAL SYNTHESIS** This three step progression is the dossier's own organising sequence.

THREE THINGS A SASB LISTING DOES NOT ESTABLISH, STATED PLAINLY

It does not establish that the topic is material for this specific company; it establishes candidacy only.	It does not establish magnitude or direction; page 7's relative test still has to run separately.	It does not substitute for the entity's own facts and circumstances judgement under IFRS Practice Statement 2.
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■ **ANALYTICAL SYNTHESIS** Stated here because SASB topic membership is the single most common shortcut this project encountered for skipping the remaining steps on pages 2, 7, and 9.

WHAT THIS PROJECT COULD AND COULD NOT VERIFY DIRECTLY

Direct access to the SASB Standards Navigator was not obtainable during this research (a JavaScript rendered navigator, not directly fetchable). Company published SASB indices were used as a workaround: Coca-Cola HBC and IBM each publish an index citing official SASB topic codes verbatim against their own disclosures.

■ **COMPANY PUBLISHED INDEX** Graded one tier below direct standard setter access throughout this dossier for that reason.

EVIDENCE CHAIN OF CUSTODY FOR SASB CLAIMS IN THIS DOSSIER



Standard setter, then a navigator that was not directly fetchable, then Coca-Cola HBC and IBM's own published indices, then this dossier, graded B, not A. ■ **ANALYTICAL SYNTHESIS** Shown to make the grading rule itself auditable.

STEP 1 OF 3

An industry topic list answers "what could plausibly matter to companies like this one," not "what matters to this one." Page 11 tests that gap directly: same industry, different exposure.

HOW DO WE COMPARE COMPANIES AND PEERS?

Same Industry Does Not Mean Same Exposure

A worked example of a peer comparison that failed the evidence bar, and what a defensible one requires.

A WORKED EXAMPLE OF A COMPARISON THAT FAILS THE EVIDENCE BAR

An early pass in this project attempted a same industry comparison between two beverage / technology peers on water related risk. One side, a beverage bottler, held up: its FY2025 Form 10-K genuinely discusses water availability, in Item 1 (Business). The other side, an initial automated read of a technology company's filing, claimed the filing "recognises water as a material operational and strategic risk," but the only supporting language was a 2030 sustainability goal and an unrelated data centre sentence.

That comparison was rejected, not repaired. Verdict retained in this dossier: no sufficiently verified company comparison found for that specific pairing. ■ **COMPARISON REJECTED**

The lesson generalises: a peer comparison is only as strong as its weaker side's evidence. One well sourced side does not rescue the other.

HETEROGENEITY WITHIN ONE INDUSTRY, SYSTEM LEVEL EVIDENCE

Among the 41 banks in the ECB's 2022 supervisory climate stress test, roughly 60% had no climate risk stress testing framework at all, and only about 20% used climate risk as a variable when granting loans, inside a single, heavily regulated industry.

ECB Banking Supervision, 2022 Climate Risk Stress Test. ■ **SYSTEM LEVEL EVIDENCE**
ANATOMY OF THE REJECTION

Side that held up. Genuine 10-K discussion of the topic, in context, page locatable, correctly graded by section, Item 1 vs Item 1A. ■ **OBSERVED FACT**

The failure mode is specific and worth naming: **weak quote overreach**, citing real text that does not actually say what it is being cited for.

KEY TAKEAWAY

Same industry, same regulatory regime, and practice still varies enormously company to company. Peer context narrows the question; it never answers it for an individual company.

WHAT A DEFENSIBLE PEER COMPARISON REQUIRES

Both companies' positions traced to primary disclosure, not one primary, one inferred.

The comparison presented as a spectrum where evidence differs in strength, not a false binary.

Any unexplained numerical discrepancy disclosed as a discrepancy, not resolved by silent selection.

Industry membership alone is never treated as proof of comparable exposure; see page 9.

Side that failed. An automated extraction cited a 2030 sustainability goal and one unrelated operational sentence as if they established a materiality position. Neither supports the claim made about it. ■ **EVIDENCE GAP**

HOW DO WE PRIORITIZE?

From a Topic List to a Prioritized Issue Set

Why a conventional materiality matrix under determines the answer, and a sequence that does not.

WHY A SIMPLE 2×2 MATRIX IS NOT ENOUGH ON ITS OWN

A conventional materiality matrix plots issues on two unlabelled or loosely defined axes, often "importance to stakeholders" vs "importance to business," and treats position in the top right quadrant as the verdict. Three problems recur.

The axes routinely blend impact materiality and financial materiality, page 3, into one score, obscuring which test an issue actually passed.

Placement is frequently a survey average, not a magnitude / nature test against the entity's own financials, page 7.

A matrix shows relative rank, not whether any issue clears the financial materiality bar at all.

A PRIORITIZED LIST IS A DATED JUDGEMENT, NOT A PERMANENT RANKING

IFRS Practice Statement 2's facts and circumstances test is applied at each reporting period, not once. A topic ranked low today can rise if the entity's exposure, the regulatory environment, or the magnitude test itself changes. Prioritization is reissued, not fixed.

THE SEQUENCE APPLIED TO A SINGLE WORKED TOPIC: WATER SCARCITY

STEP	APPLIED TO A BEVERAGE BOTTLER IN A STRESSED BASIN
1. Identify	Water is a listed SASB topic for the beverage industry.
2. Nature / magnitude	Basin stress could reasonably affect production volume and input cost, page 2's illustrative contrast.
3. Exposure screen	Facility is physically located in a designated water stressed basin, page 9's first filter.
4. Rank	High magnitude, high likelihood, near term horizon, ranks near the top of the survivor set.

■ **ILLUSTRATIVE** This worked walk through uses the illustrative contrast already introduced on page 2; it is a demonstration of the sequence's mechanics, not a new disclosed company case.

A PRIORITIZATION SEQUENCE GROUNDED IN THE FRAMEWORK ALREADY BUILT

- 1

Start from the SASB informed candidate list.
Page 10
- 2

Apply the IFRS PS2 nature / magnitude test to each.
Page 2, relative to this entity
- 3

Screen by exposure, vulnerability, resilience.
Page 9, company specific
- 4

Rank survivors by magnitude × likelihood × horizon.
Page 8

This four step sequence is the dossier's own synthesis, built from the components already sourced on pages 2, 7, 8, and 10. ■ **ANALYTICAL SYNTHESIS**

WHAT A MATRIX IS STILL USEFUL FOR

As a communication aid summarising a prioritization already reached through the sequence above, not as the analytical method that reaches it.
■ **ANALYTICAL SYNTHESIS**

MODULE 4 CONCLUSION

Prioritization is the output of applying the earlier tests in sequence to one company's facts, not a separate scoring exercise. Module 5 now asks what a company actually does once an issue survives this filter.

WHAT CAN MANAGEMENT DO?

From Risk Identification to Management Response

Once an issue clears the materiality bar, what can management actually do about it?

A RESPONSE TAXONOMY, LABELLED AS DOSSIER SYNTHESIS

Avoid / exit. Cease the activity or exposure entirely.

Reduce. Lower the probability or severity of the effect.

Adapt. Change operations to perform acceptably despite the exposure.

Transform. Redirect strategy so the issue becomes a source of advantage.

Accept / retain. Absorb the exposure where mitigation cost exceeds benefit.

This five category taxonomy is developed for this dossier. ■ **ANALYTICAL SYNTHESIS** It is not attributed to IFRS, ISSB, COSO, or ISO. A general risk management vocabulary, avoid / reduce / transfer / accept, exists, but this project could not verify a specific ISO 31000 mapping to this exact five part form, and none is claimed.

WHERE THE OTHER THREE CATEGORIES ALREADY APPEARED IN THIS DOSSIER

Reduce

UPS ORION: lowering fuel / mileage exposure through operating technology, page 6.

Transform

Neste's Renewable Products pivot attempted this; the 2024 evidence did not sustain the claim, page 15.

Accept / retain

Any topic that fails the page 12 prioritization sequence is, by construction, being accepted without active response.

WHY RESPONSE IS GRADED HERE, NOT JUST DESCRIBED

Naming a response category is not the same as showing it worked. Avoid / exit and adapt / continue are both defensible choices in principle; which one was right for Ørsted's two projects is a capital allocation question, not a taxonomy question, which is exactly what page 14 tests next.

ØRSTED, ONE COMPANY, TWO RESPONSES, SAME QUARTER

In October 2023, Ørsted applied two different responses from the taxonomy to two different US projects simultaneously.

Ocean Wind 1 & 2, avoid / exit. Development ceased; DKK 19.9bn of the FY2023 impairment relates to Ocean Wind 1 alone; a separate DKK 8 to 11bn provision was estimated for cessation costs.

Revolution Wind, adapt / continue. Final investment decision taken the same day, disclosed as offering "a forward looking spread to WACC above Ørsted's guided range," continued investment, but only once that hurdle was met.

■ **OBSERVED FACT** Ørsted, nine month 2023 interim announcement (24 Oct 2023).

KEY TAKEAWAY

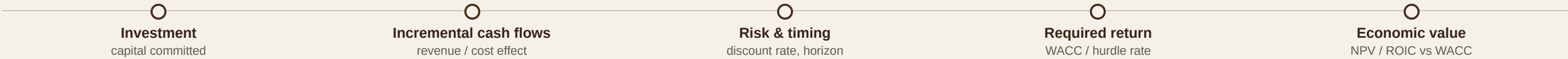
Management response is not one company wide posture. The same company held a strategic direction, offshore wind, while making opposite project level decisions in the same announcement, which is why response has to be evaluated project by project, not company by company.

WHAT DOES THE RESPONSE COST?

Cost, or Investment?

What the response requires in capital, and the same NPV / ROIC test every capital allocation decision must clear.

THE FINANCIAL CHAIN A RESPONSE MUST CLEAR



A management response is a capital allocation decision like any other: it is only value creating if the risk adjusted return on the capital committed exceeds the cost of that capital. Spending money on a response is not, by itself, evidence of value creation. ■ **OBSERVED FACT**

Mauboussin & Callahan, ROIC / WACC framework.

THREE RESPONSE TYPES BY CAPITAL CHARACTER, DOSSIER SYNTHESIS

- Defensive.** Capital spent to avoid a downside, compliance capex for example; success is measured against the avoided loss, not a positive return.
- Transitional.** Capital redeploying an existing asset base toward a changed operating environment; Ørsted's continued Revolution Wind investment is transitional in character.
- Growth.** Capital seeking a new value pool the issue has created; evaluated exactly like any other growth project, against WACC.

■ **ANALYTICAL SYNTHESIS** Category labels developed for this dossier; the underlying NPV / ROIC test above is not.

SAME NUMBER, DIFFERENT QUESTION FROM PAGE 7

Page 7 used –14% / 12.9% to illustrate magnitude, how large, relative to what. Here the same two figures are used to test value creation, whether return cleared the cost of capital. Same evidence, different analytical question.

KEY TAKEAWAY

ESG related spending is neither automatically a cost nor automatically value creation. It is a capital allocation decision, tested the same way every capital allocation decision is tested, and page 15 now runs that test to a conclusion.

DID IT CREATE ECONOMIC VALUE?

From ESG Strategy to Economic Value

Did the decision actually create value? Three real cases, tested with the same discipline, three different answers.

POSITIVE CASE: UPS ORION

>\$400m

Annual cost savings, UPS reported

\$250m

Full deployment cost, third party estimate

Management claim vs analyst test: UPS's own 2016 Annual Report states ORION "is now generating more than \$400 million in annual cost savings and avoidance," alongside the disclosed operating fact that miles driven rose only 0.2% against 4%+ volume growth, a mechanism, not just an assertion.

- **OBSERVED FACT** the \$400m figure and mileage data.
- **THIRD PARTY ESTIMATE** the \$250m investment figure, INFORMS, not UPS's own filing; figures are a decade old.

THREE CASES, THREE VERDICTS, SIDE BY SIDE

CASE	CHANNEL	VERDICT	BASIS
UPS ORION	Cost	Value created	Disclosed mechanism plus disclosed financial effect, both observed fact
Ørsted US Offshore	Asset / capex	Mixed, US specific	Audited loss is fact; the "not a general failure" reading is analyst inference
Neste Renewable Products	Revenue / margin	Set aside	One year swing traced substantially to a tax credit expiry, not a durable edge

DIFFICULT CASE: ØRSTED

Management claim. The FY2023 results announcement is headlined "strong underlying results and strategic progress in a challenging year." ■ **MANAGEMENT CLAIM**

Independently observable fact. DKK 26.8bn audited impairment; ROCE −14% including it, 12.9% excluding it.

■ **OBSERVED FACT**

Analyst inference, not a disclosed fact. The impairment reflects identifiable US market conditions rather than a general failure of offshore wind economics; continued disciplined investment in Revolution Wind under an explicit WACC spread test supports that reading. ■ **ANALYST INFERENCE**

CASE CONSIDERED AND SET ASIDE

Neste's Renewable Products segment was an early candidate: comparable return on net assets was 18.9% in 2023 but collapsed to −0.9% in 2024, with comparable EBITDA falling from €1,906m to €514m, substantially explained by the expiry of the US Blender's Tax Credit, €590m contribution in 2024, rather than an isolable strategic edge.

- **OBSERVED FACT** Neste, FY2024 Financial Statements Release. Set aside as the flagship case: one strong year does not establish a durable strategic edge.

MODULE 5 CONCLUSION

The same framework, applied honestly, produces a clean positive case, a genuinely mixed case, and a case that does not survive scrutiny at all, which is the point: the framework is built to reach whichever of the three the evidence actually supports.

WHAT IF THE FUTURE DOES NOT FOLLOW THE BASE CASE?

The Future Is Not One Forecast

Why sustainability related financial analysis must work across multiple plausible states of the world.

FORECAST AND SCENARIO ARE DIFFERENT OBJECTS

Forecast. A single best estimate path, typically with a confidence interval around one most likely trajectory. Answers: "what do we expect to happen?"

Scenario. An internally consistent, named alternative state of the world, one of several, deliberately not ranked by probability. Answers: "how would our strategy hold up if this happened instead?"

SHELL, 1973: SCENARIOS USED, NOT PREDICTED

"Neither rigid predictions nor wild fantasies, building scenarios is a craft that holds real commercial value."

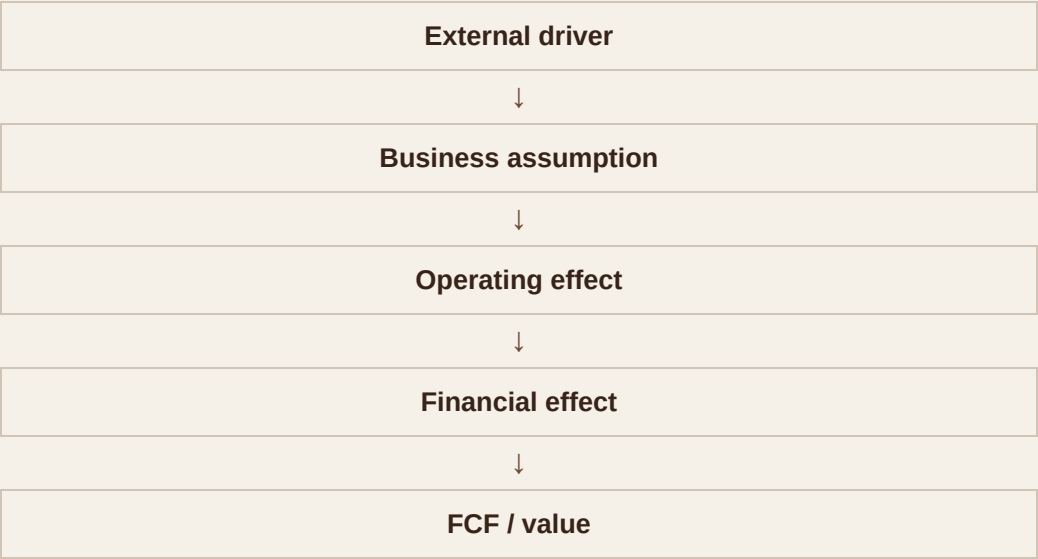
Shell's planning team had, by September 1972, built scenarios anticipating conditions resembling Middle East oil disruption. When the October 1973 crisis occurred, the company states its "decision makers were mentally prepared for the worst because they had already imagined it," crediting the scenario work with "financial gains running into the billions of dollars" from decisions on refinery sales and non replacement.

■ **COMPANY REPORTED RETROSPECTIVE** Shell, "40 Years of Shell Scenarios." The financial benefit figure is Shell's own qualitative characterisation, not an audited line item, shown as such, not as independently verified causal proof.

WHY THIS DOSSIER DOES NOT STOP AT FORECASTING

A single point forecast, however carefully built, answers only "what do we expect." It is silent on what happens to the strategy under discussion if the world instead moves toward one of the other plausible states described on page 8, orderly, disorderly, hot house, or too little too late. Scenario work exists specifically to close that gap, which is why Module 6 builds a scenario, pressure tests it against a company's numbers on page 17, and only then asks what management should actually do about it on page 18.

THE CONSTRUCTION CHAIN THIS DOSSIER USES



■ **ANALYTICAL SYNTHESIS** No single standard setter publishes this exact five step chain. It combines IFRS S2 / TCFD scenario categories with standard corporate finance identities, and is labelled as this dossier's own construction throughout.

KEY TAKEAWAY

Building one scenario is storytelling; building several and testing a strategy against all of them is analysis. Page 17 now pushes one scenario all the way through a company's numbers to see what actually breaks.

WHAT BREAKS FIRST?

Stress Testing the Business Model

Tracing external scenarios through operating and financial statements, with precise, non synonymous terminology.

TERM	WHAT CHANGES	EBA / GL / 2018/04 DEFINITION, VERBATIM
Sensitivity analysis	One risk factor changes, others held constant	"a stress test that measures the potential impact of a specific single risk factor"
Scenario analysis	A coherent, internally consistent set of risk factors changes together	"a set of risk factors ... aligned in an internally consistent way ... simultaneous ... forward looking"
Stress test	Impact of a scenario on the overall capital / financial position	"the assessment of the impact of certain developments ... on the overall capital position"
Reverse stress test	Start from a failure outcome, work backward to its causes	"starts from the identification of the pre defined outcome ... and then explores scenarios ... that might cause this to occur"

■ **OBSERVED FACT** European Banking Authority, Guidelines on Institutions' Stress Testing, EBA/GL/2018/04, 19 Jul 2018, §2 Definitions. Written for regulated banks under EU law; applying "reverse stress test" to a general corporate strategy context is this dossier's own extension of the concept. ■ **ANALYTICAL SYNTHESIS**

SYSTEM LEVEL EVIDENCE OF SCENARIO TO FINANCIAL TRANSMISSION

The ECB's 2022 climate stress test ties a named transmission channel to a quantified system wide outcome: physical risk "translates through a decrease in sectoral productivity ... and an increase in loan losses in the affected areas," producing an aggregate €70bn credit / market loss across 41 banks.

■ **SYSTEM LEVEL EVIDENCE** ECB Banking Supervision, 2022 Climate Risk Stress Test. This is system level evidence across 41 institutions, not a single company case; no comparably precise single company, named scenario, quantified financial impact disclosure was located in this project. One large UK bank's 2024 TCFD report states plainly it has not performed scenario analysis on its assets under management. ■ **EVIDENCE GAP**

ILLUSTRATIVE STRESS TEST, NOT COMPANY DATA

METRIC	BASE	STRESS	DRIVER
Revenue	100	92	Volume / price effect
EBIT margin	15%	12%	Cost inflation partly offsets
EBIT	15.0	11.0	Revenue × margin
Capex	8	11	Adaptation spend rises

■ **ILLUSTRATIVE** All figures on this table are illustrative constructs to show the arithmetic, not a real company's results.

KEY TAKEAWAY

The chain from scenario to a number that moves is traceable, at system level with real figures, and at company level with an honestly labelled illustration. Page 18 asks what management should actually do once the stress test is run.

WHAT SHOULD MANAGEMENT DO UNDER UNCERTAINTY?

From Scenario to Strategic Decision

Using uncertainty to decide whether to invest, wait, adapt, hedge, or exit.

ROBUST DOES NOT EQUAL OPTIMAL IN ONE FUTURE

A robust strategy is defined against a wide range of futures, not against the single most likely one: it "performs well over a wide range of futures and better manages surprise," explicitly not the strategy that maximises performance in any one scenario.

■ **OBSERVED FACT** RAND Corporation, Robust Decision Making research brief, drawing on Lempert, Groves, Popper & Bankes, *Management Science* (2006).

Invest

Adapt

Hedge

Wait / stage

Exit

Five response options, dossier synthesis.

WHERE THE EVIDENCE FOR EACH RESPONSE OPTION ACTUALLY STANDS

Invest	Adapt	Hedge	Wait / stage	Exit
Standard NPV / ROIC test, page 14.	Ørsted's Revolution Wind continuation, page 13.	Southwest case, left.	Dixit & Pindyck; Maersk sequence, left.	Ørsted Ocean Wind 1/2, page 13.
■ OBSERVED FACT	■ OBSERVED FACT	■ OBSERVED FACT	■ ANALYST INFERENCE	■ OBSERVED FACT
■ EVIDENCE GAP "Adapt" as a category distinct from the other four was not independently source verified in this project and is used descriptively, not as a term of art.				

KEY TAKEAWAY

Under uncertainty and irreversible investment, the ability to wait, learn, scale, or abandon may itself have economic value, verified in Dixit & Pindyck (1994) and applied directly to sustainability investment in Han Smit, "Sustainability Real Options," California Management Review (2025). This is not a claim that waiting is always optimal; its value depends on uncertainty, irreversibility, timing, and opportunity cost.

Sources: [RAND / Lempert et al. — Robust Decision Making \(2006\)](#) · [Maersk — press releases \(2021, 2024\)](#) · [Southwest Airlines — Form 10-Q \(Q3 2025\)](#) · [Dixit & Pindyck \(1994\)](#) · [Smit \(2025\)](#)

MAERSK, STAGED INVESTMENT LOGIC

8 methanol dual fuel vessels ordered 2021, plus a 4 vessel option, growing to a 20 vessel dual fuel order by December 2024, alongside Maersk's own acknowledgement that "sourcing an adequate amount of carbon neutral methanol from day one ... will be challenging."

■ **OBSERVED FACT** Maersk press releases, Aug 2021 & Dec 2024. The sequence is consistent with staged investment / strategic flexibility logic; Maersk itself does not describe the strategy in real options terms.

■ **ANALYST INFERENCE**

SOUTHWEST, HEDGING IS A DECISION, NOT A PERMANENT STATE

Southwest's fuel hedging, from 1994, is company reported to have saved "an estimated \$3.5 billion" 1998 to 2008. ■ **COMPANY REPORTED** In its Q3 2025 Form 10-Q, Southwest disclosed it discontinued the program, citing "higher fuel hedging premium costs over time." ■ **OBSERVED FACT**

The lesson is not that hedging is good or bad, it is that the economics of a risk management mechanism can change and be revisited.

HOW DOES AN ANALYST INTEGRATE EVERYTHING?

From Analysis to Investment Decision

Evidence hierarchy, fact vs claim vs inference, and one complete case run through the entire workflow.

EVIDENCE QUALITY HAS TWO DIMENSIONS

Appropriateness, quality: relevance and reliability, and sufficiency, quantity, are independent, and reliability "is influenced by its source and its nature."

■ **OBSERVED FACT** IAASB, ISA 500, §5(b),(e). Grounds financial audit evidence, not ESG research directly, used here as the strongest available analogy for this dossier's own A to E evidence grading, not as a claim of audit compliance.

■ **ANALYTICAL SYNTHESIS**

FACT / CLAIM / INFERENCE, FORMALLY GROUNDED

US securities law distinguishes forward looking statements, "believes," "expects," "plans," from historical fact, requiring the former to carry cautionary language.

■ **OBSERVED FACT** SEC / Private Securities Litigation Reform Act 1995 safe harbor framework. CFA Institute Standard V(A) separately requires "a reasonable and adequate basis ... for any investment analysis, recommendation, or action."

■ **OBSERVED FACT**

THE ANALYST WORKFLOW THIS DOSSIER FOLLOWS

Identification → materiality → financial effect → risk / uncertainty → management response → capital allocation → valuation / decision. ■ **ANALYTICAL SYNTHESIS** No single external framework publishes these seven steps as one workflow; each step traces individually to pages 1 to 18.

INTEGRATED DECISION MATRIX, DOSSIER SYNTHESIS

EVIDENCE	FINANCIAL IMPACT	UNCERTAINTY	INDICATED RESPONSE
Strong	High	Low	Act
Strong	High	High	Stage / hedge
Moderate	Moderate	High	Investigate further
Weak	Unclear	High	Do not overclaim

■ **ANALYTICAL SYNTHESIS** Quadrant labels are dossier synthesis, each individually grounded: CFA V(A), Dixit & Pindyck / RAND, ISA 500.

COMPLETE CASE, ØRSTED, INTEGRATED

Materiality	Offshore wind is core, not peripheral ■ ANALYST INFERENCE
Exposure	US more exposed than EU portfolio ■ OBSERVED FACT
Mgmt response	Ocean Wind cancelled; Revolution Wind FID taken ■ OBSERVED FACT
Capital allocation	DKK 26.8bn impairment (vs 28.4bn interim, unreconciled) ■ OBSERVED FACT
Financial outcome	ROCE −14% incl., 12.9% excl. ■ OBSERVED FACT
Management claim	"Strong underlying results" headline ■ MANAGEMENT CLAIM
Analyst judgement	US specific, not proof of general failure ■ ANALYST INFERENCE

Evidence gaps, stated rather than filled: the two impairment figures do not reconcile; no project level IRR / ROIC was ever disclosed for any of the three named projects.

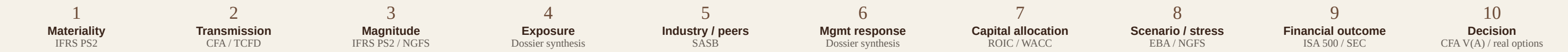
KEY TAKEAWAY

This case can be compressed to one page without becoming superficial because its gaps are documented as carefully as its facts, which is the discipline page 20 now generalises into a complete framework.

WHAT IS THE COMPLETE FRAMEWORK?

From Sustainability Issue to Financial Decision

A practical framework for analysing when ESG becomes economically material.



■ **ANALYTICAL SYNTHESIS** The ten stage chain above is analytical synthesis developed for this dossier. Each stage independently traces to a source shown beneath it and on pages 1 to 19; the chain itself is not published by IFRS, ISSB, CFA Institute, or any other body as one framework.

REMEMBERED AS FIVE QUESTIONS

- 1. **Does it matter?** Financial materiality
- 2. **How does it matter?** Transmission mechanism
- 3. **How much does it matter?** Magnitude, uncertainty, exposure, horizon
- 4. **What will management do?** Response and capital allocation
- 5. **What should we conclude?** Scenario tested judgement



HOW TO USE THIS FRAMEWORK ON A NEW TOPIC

Start at stage 1 with the entity's own facts, not an industry average or an ESG score.	Do not skip a stage because an earlier one looked conclusive; magnitude without exposure, or response without a capital test, is an incomplete answer.	Grade every claim as it is made, fact, claim, or inference, rather than retrofitting the grading at the end.
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WHAT THIS FRAMEWORK DOES NOT DO

Equate ESG scores with financial value. Assume every issue is material. Predict the future. Treat management claims as realized outcomes. Assume ESG investment automatically creates value.

Replace company analysis with an industry label. Eliminate uncertainty. Produce mechanically correct decisions. Guarantee a good strategy returns positively. Replace professional judgement.

FINAL THESIS

"The objective is not to prove that ESG creates value. It is to determine when, through what mechanism, under which assumptions, and with what financial consequences a sustainability related issue changes, or does not materially change, the economics of a company."

Sources: synthesis of pages 1 to 19 sources, principally IFRS S1 / S2, IFRS Practice Statement 2, CFA Institute, EBA/GL/2018/04, NGFS, ISA 500, SEC / PSLRA, Dixit & Pindyck (1994), Smit (2025). See the Source Register, page 21, for the complete list with links.

WHERE DID EVERY CLAIM COME FROM?

Source Register

Every citation used across pages 1 to 20, graded and linked. A hyperlink appears only where an official URL was directly verified; nothing below is invented.



DOCUMENT	DATE	GRADE	USED ON
IFRS S1, General Requirements — IFRS Foundation	2023	A	1, 2, 10
IFRS S2, Climate-related Disclosures — IFRS Foundation	2023	A	8, 16
IFRS Practice Statement 2, Making Materiality Judgements	2017	A	2, 7, 12
The Coca-Cola Company, FY2025 Form 10-K	2026	A	1
SEC Staff Accounting Bulletin No. 99	1999	A	2
Private Securities Litigation Reform Act, safe harbor	1995	A	19
ISSB / EC / EFRAG, Interoperability Guidance	May 2024	A	3
SASB Standards — IFRS Foundation	ongoing	A	3, 10
Coca-Cola HBC, published SASB Index	2022	B	3, 10
IBM, published SASB Index	2023	B	3, 10
CFA Institute, ESG Integration Guidance	Nov 2022	A	4
CFA Institute, Standard V(A)	ongoing	A	19
TCFD, Technical Supplement, Fig. 3	Jun 2017	A	4, 5
IFRS Conceptual Framework	2018	A	5
UPS, 2016 Annual Report	2016	A	6, 15
INFORMS, 2016 Franz Edelman Award case study	2016	C	6, 15

DOCUMENT	DATE	GRADE	USED ON
Ørsted, FY2023 Annual Report announcement	Feb 2024	A	6, 7, 9, 13-15, 19
Ørsted, Nine Month 2023 interim announcement	Oct 2023	A	6, 9, 13
ECB Banking Supervision, 2022 Climate Risk Stress Test	Jul 2022	A	7, 11, 17
NGFS, Climate Scenarios Technical Documentation, Phase V	Nov 2024	A	8, 16
EBA, Guidelines on Institutions' Stress Testing	Jul 2018	A	17
Mauboussin & Callahan, Return on Invested Capital	2022	A	14
Neste, FY2024 Financial Statements Release	Feb 2025	A	15
Shell, "40 Years of Shell Scenarios"	2013	B	16
RAND / Lempert et al., Robust Decision Making	2006	A	18
Maersk, press release, 8 methanol vessels	Aug 2021	A	18
Maersk, press release, 20 dual-fuel vessels	Dec 2024	A	18
Southwest Airlines, Form 10-Q	Q3 2025	A	18
Southwest Airlines, hedging retrospective (\$3.5bn)	n.d.	B	18
Dixit & Pindyck, Investment Under Uncertainty	1994	A	18
Smit, Sustainability Real Options, Calif. Mgmt Review	2025	A	18
IAASB, ISA 500, Audit Evidence	rev. 2022	A	19

GRADING KEY

A: direct primary source: standard setter, regulator, or the subject company's own filing / announcement. **B:** company published derivative (an index or retrospective built from, but not itself, the primary filing). **C:** third party estimate, not produced by the subject company. Two sources referenced in the dossier text without an independently verifiable public URL (a UK bank's 2024 TCFD report, cited for its own disclosed evidence gap on page 17) are described in the running text rather than listed here, per this dossier's rule against inventing a citation link.